Driver Overtake Interactions in Formula One

Data Mining (INFSCI 2160) | Spring 2026 | Monika Herbst

Motivation & Goals:

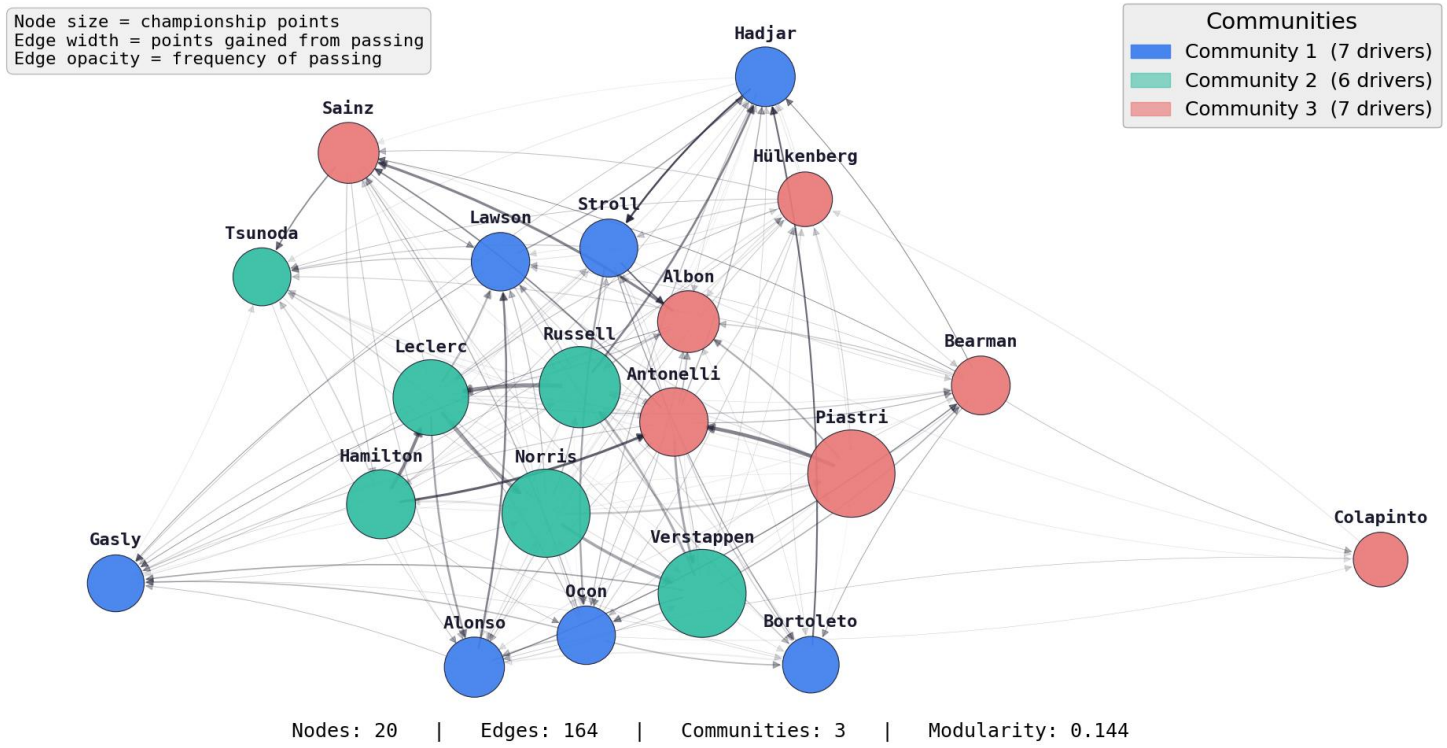
- Analyze driver overtakes in F1 using network approach:
  - Allows for deeper understanding of driver interactions
  - Network visualizations provide intuitive illustration of how drivers interact
- Determine if network is fully connected
- Assess if clustering behavior is present
- Identify groups of drivers using community detection (Louvain algorithm)
  - Show communities in network visualization
  - Compare with final standings to see if they are related

Data & Network Summary:

- **Dataset (Kaggle)**: Formula 1 Race Data (jtrotman)
  - Comprehensive lap-by-lap data
  - Usable data available from 1996-present
- **Time period analyzed**: 2025
  - 24 grand prix (full) races
  - 6 sprint (mini) races
- **Nodes**: drivers
- **Edges**: net interactions
  - Overtakes identified using lap-by-lap data
  - The “value” of an overtake corresponds with the number of points the change in position is worth
  - Edges are weighted and directed (no bidirectional edges)
- **Removed**: disqualifications and drivers with <10 races

Results & Network Visualization

F1 2025 Driver Overtake Network:



Final Championship Standings:

Driver	Total Points	Community
Lando Norris	423	2
Max Verstappen	421	2
Oscar Piastri	410	3
George Russell	319	2
Charles Leclerc	242	2
Lewis Hamilton	156	2
Kimi Antonelli	150	3
Alexander Albon	73	3
Carlos Sainz	64	3
Fernando Alonso	56	1
Nico Hulkenberg	51	3
Isack Hadjar	51	1
Oliver Bearman	41	3
Liam Lawson	38	1
Esteban Ocon	38	2
Lance Stroll	33	1
Yuki Tsunoda	33	1
Pierre Gasly	22	1
Gabriel Bortoleto	19	1
Franco Colapinto	0	3

Results:

- Network is fully connected | Clustering behavior is present and extreme
- Average path length: 1.59 | Average clustering: 0.905
- Three groups of drivers found using community detection (Louvain algorithm)
  - Low modularity => not that significant
  - Somewhat related to final standings but not perfect: 1 – back of the field | 2 – front of the field | 3 – middle of the field